YAKEEN

NEET 2.0 2027

Physics | Lecture - 13

Motion in a Straight Line

By- Manish Raj (MR Sir)



Todays goal

H/W goal

→ S.A of 1-D → 6

all case of motion under gravity

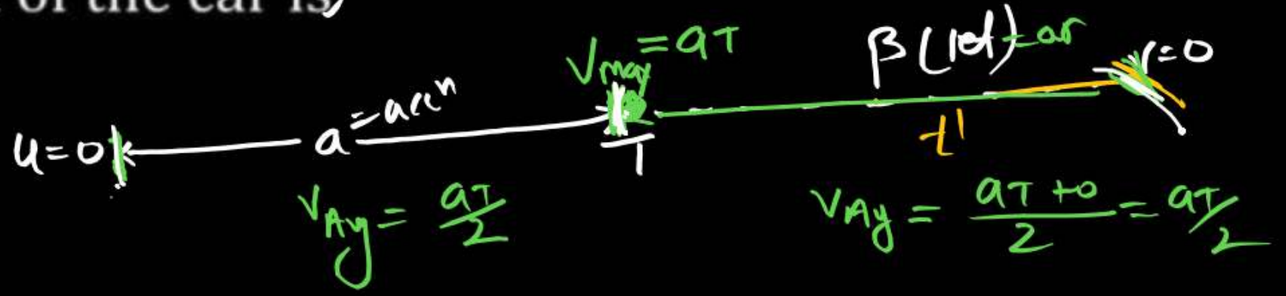
- Ground to ground
- Juggle
- Projected from some height
- Balloon drop
- Rocket Prop.

Question

H/w kal discuss (copy) H/w kal karke aana

A car starts from rest and moves with uniform acceleration a on a straight road from time $t = 0$ to $t = T$. After that, a constant deceleration brings it to rest. In this process the average speed of the car is

1. aT
2. $\frac{aT}{4}$
3. $\frac{3aT}{2}$
4. $\frac{aT}{2}$



Avg speed = Avg velocity $= \frac{aT}{2}$

Motion under gravity

Motion under gravity

↓ Special case of motn with cost^n accn.

→ Motion with Uniform acceleration

→ Non-Uniform motion, Variable Velocity

→ $a = g$ (air resistance ignore)

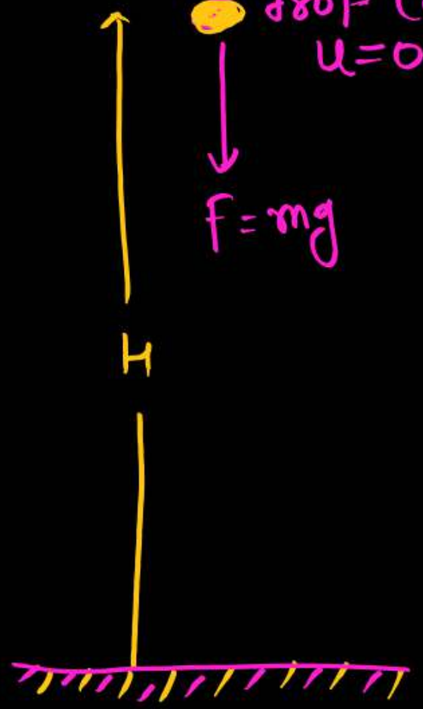
→ Jabtak question me air resistance given nahi ho use consider Nahi Karna hai

→ all eqn of motion is Valid here ✓

→ all case of consn accn is Valid here ✓

→ In question accn is not given then also consn $a = g = 10 m/s^2$

Case-1



$a = \frac{F}{m} = \frac{[mg]}{m}$

$a = g = 9.8 m/s^2 = cost^n$

$\simeq 10 m/s^2$ ✓

= does not depend on mass

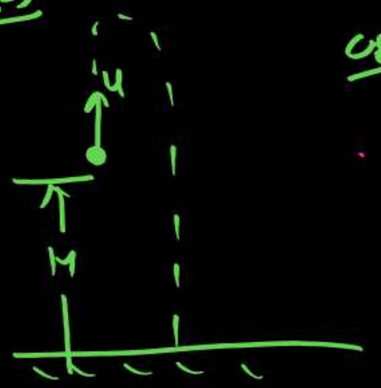
$S = H = \frac{1}{2} g t^2$

Same

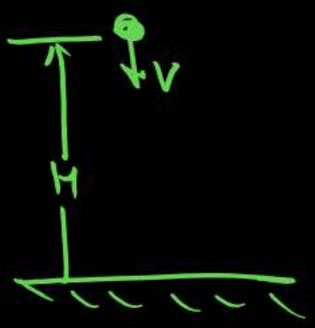
Case-2



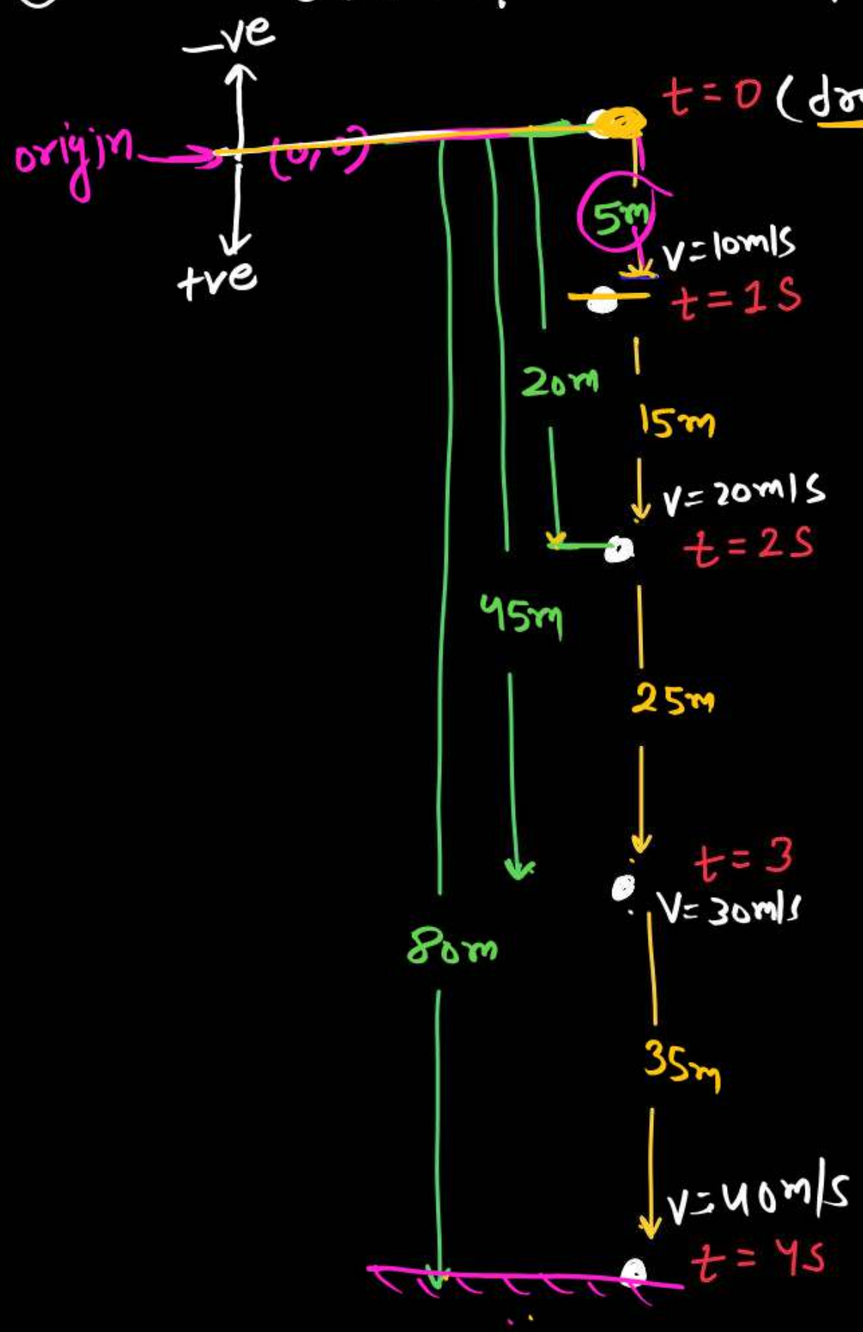
Case-3



Case-4



① object is droped ($u=0$)



$a = g$ [downward]

$a = +g = +10\ m/s^2$

for displacement in 't' sec

$s = ut + \frac{1}{2}at^2$

$s = \frac{1}{2}g(t)^2$ ← Parabola.

$S_{1s} = \frac{1}{2}10(1)^2 = 5m$ ×

$S_{2s} = \frac{1}{2}10(2)^2 = 20m$ = 15m

$S_{3s} = \frac{1}{2}10(3)^2 = 45m$

$S_{4s} = \frac{1}{2}10(4)^2 = 80m$

$S_{5s} = \frac{1}{2}10(5)^2 = 125m$

$S_{6s} = 180m$

$S_{1s} : S_{2s} : S_{3s} : S_{4s} : S_{5s} = 5 : 20 : 45 : 80 : 125$

$= x : 4x : 9x : 16x : 25x$

$S_{1st} : S_{2nd} : S_{3rd} : S_{4th} : S_{5th} = 5 : 15 : 25 : 35 : 45$

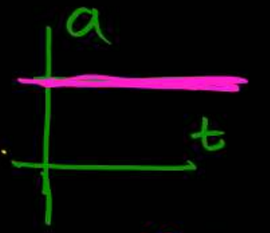
$= x : 3x : 5x : 7x : 9x$

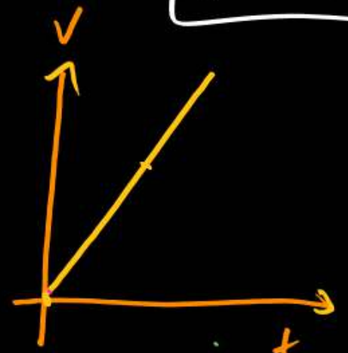
ratio of odd no

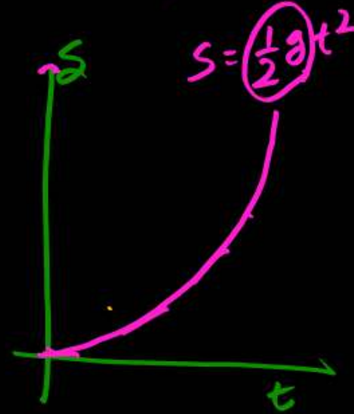
$v = u + at$ (velocity at t)

$v = 0 + gt$

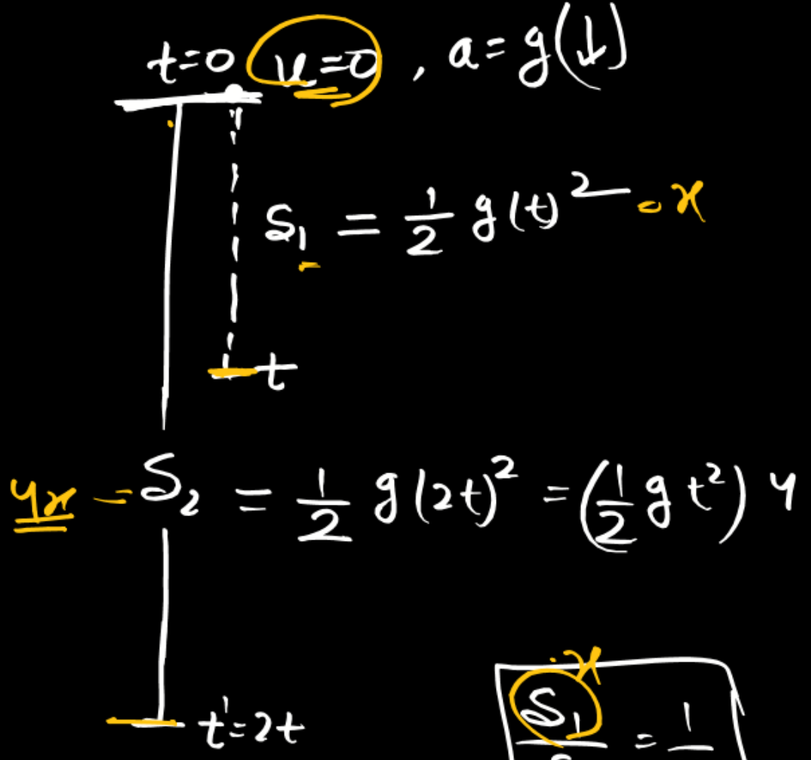
$v = 10t$







(2) object is dropped then find ratio of displ^m in time t & '2t'



$$\frac{S_1}{S_2} = \frac{1}{4}$$

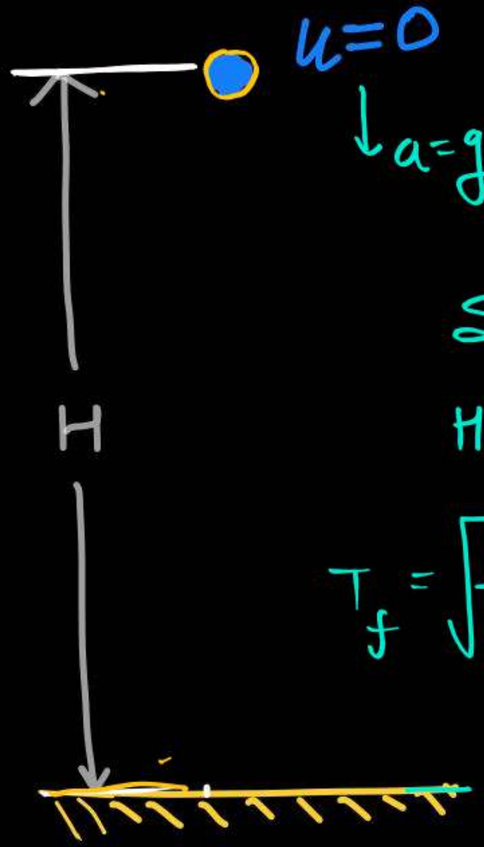
$S_2 = 4S_1$

Wake up

$S_t : S_{next\ t} = x : 3x$

(3)

Object is droped from height 'H' then find time taken to reach ground and velocity at ground.



$s = ut + \frac{1}{2}at^2$ ✓

$H = \frac{1}{2}gt^2$

$T_f = \sqrt{\frac{2H}{g}}$

$u_i = 0$
$a = g$
$s = H$
$v_f = ??$

3rd eqn of motion

$v^2 - u^2 = 2as$

$v^2 = 2gH$

$v = \sqrt{2gH}$ ✓

(5)

Ball is droped then find ratio of distance in 3rd sec and 7th sec

displament same here

Soln

$$S_{3rd\ sec} = u + \frac{a}{2}(2n-1)$$

$$S_{3rd} = 0 + \frac{10}{2}(2\times 3 - 1)$$

$$= 5(5) = 25m$$

$$S_{7h} = 0 + \frac{10}{2}(2\times 7 - 1)$$

$$= 65m$$

$$\frac{S_{5th}}{S_{7th}} = \frac{25}{65} = \frac{5}{13}$$

methad -2

ratio = 1 : 3 : (5) : 7 : 9 : 11 : (13) : 15

$= \frac{5}{13}$ ✓

(9) Ball is droped from some height and travel 55m in last-sec of its Journey then find time of flight and max^m height.

$S_{nth} = u + \frac{a}{2}(2n-1)$

$55 = 0 + \frac{10}{2}(2n-1)$

$11 = 2n-1$

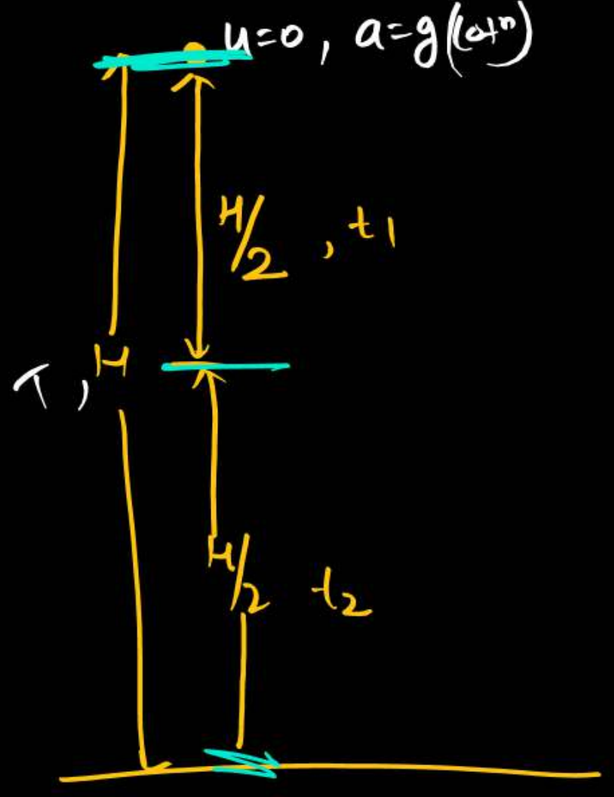
$12 = 2n$

$n = 6\,sec$

$S_{nth} = \frac{10}{2}(2n-1)$

$S_n = 10(n) - 5$

(10) Ball is droped from Height H then find ratio of time in 1st half and 2nd half distance.



$t_1 : t_2 = 1 : \sqrt{2}-1$ sec done ✓

$t_1 = \frac{T}{\sqrt{2}}$ —— done ✓

$t_2 = T - \frac{T}{\sqrt{2}}$

$= T\left(1 - \frac{1}{\sqrt{2}}\right)$

$= T\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$ ✓

(11) Object droped & it takes 1 sec to cover 2nd half height; then find time of flight.

$$t_2 = T\left(1-\frac{1}{\sqrt{2}}\right)$$

$$1 = T\left(1-\frac{1}{\sqrt{2}}\right)$$

$$1 = T\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$T = \left[\frac{\sqrt{2}}{\sqrt{2}-1}\right] \text{sec}$$

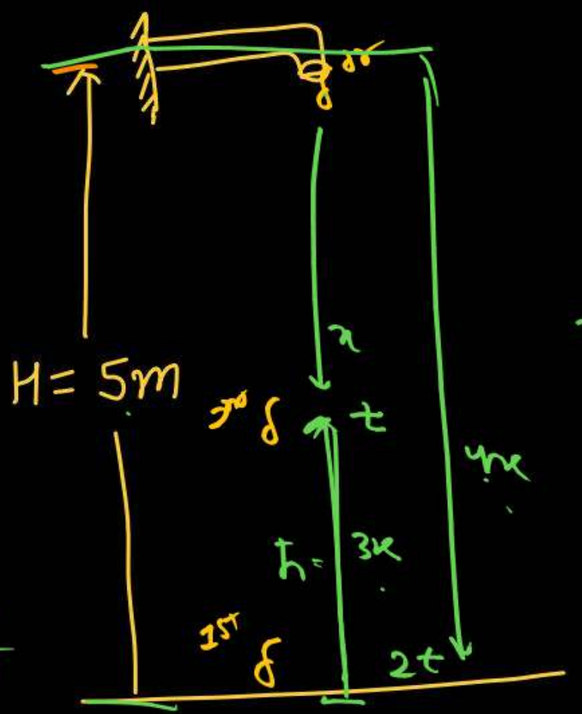
$$T = \frac{\sqrt{2}}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \frac{\sqrt{2}(\sqrt{2}+1)}{(\sqrt{2})^2-(1)^2}$$

$$= \frac{2+\sqrt{2}}{1}$$

$$= (2+\sqrt{2})$$

(12) Water drop is falling in regular intraval then find height of 2nd drop from ground if 3rd drop is about to fall when 1st just reach ground.



t = time

$4x = 5$

$x = \frac{5}{4}$

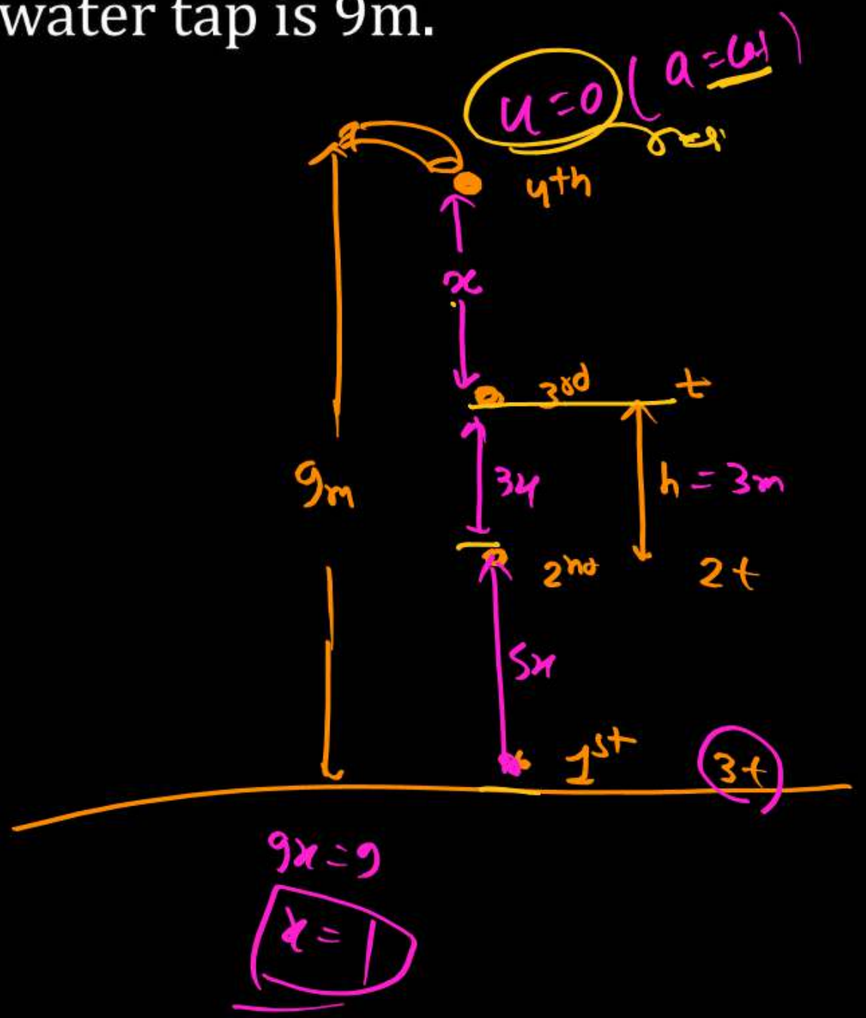
$h = 3x$

$= 3\left[\frac{5}{4}\right] = \frac{15}{4} = 3.75$

Question

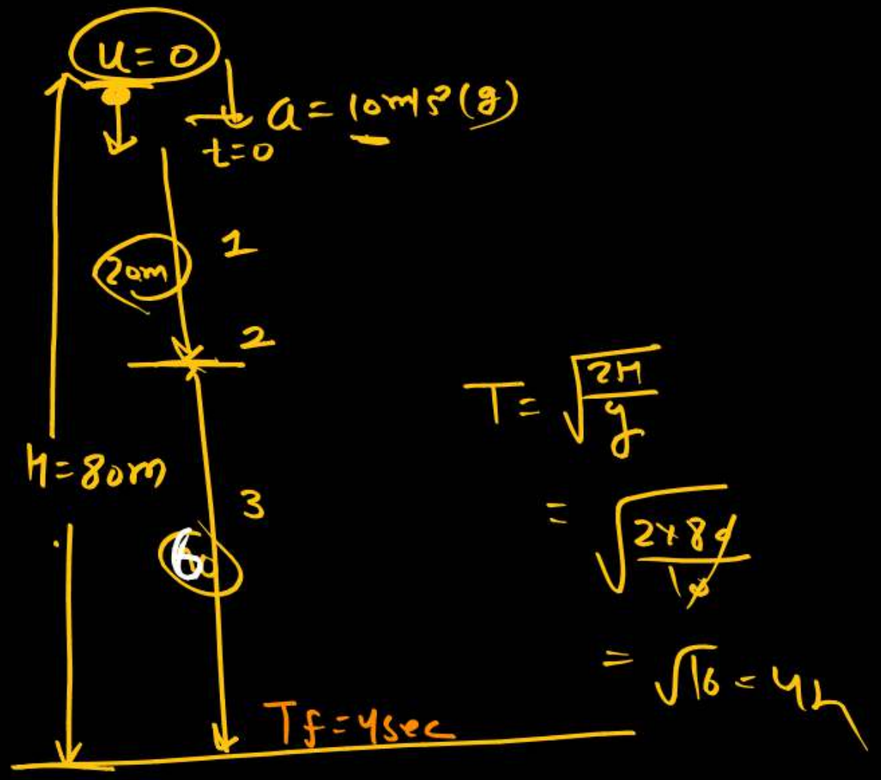
(13)

Water drop is falling in a regular intervals when 1st drop is reaches to ground then 4th drop is about to release, then find distance between 2nd drop and 3rd drop. Height of water tap is 9m.



1/0

(14) Ball is droped from 80m then find ratio of distance moved in 1st 2sec and last 2-sec of Journey.



$$T = \sqrt{\frac{2H}{g}}$$

$$= \sqrt{\frac{2\times 8\cancel{0}}{\cancel{10}}}$$

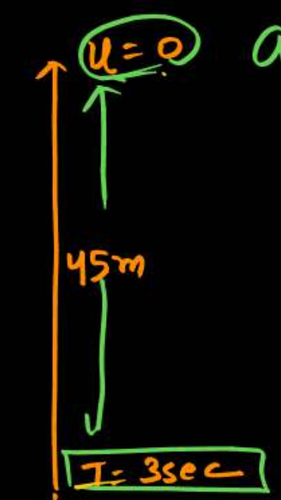
$$= \sqrt{16} = 4s$$

$$\frac{h_1}{h_2} = \frac{2\cancel{0}}{6\cancel{0}} = 1/3$$ ✓

(15)

Object is droped and distance in last 1sec is equal to 1st 3sec then find height from ground from where ball is droped

Copy



$a = 10 m/s^2$

$s = ut + \frac{1}{2}at^2 = \frac{1}{2}(10)(3)^2 = 45m$

$S_{nth} = u + \frac{a}{2}(2n-1)$

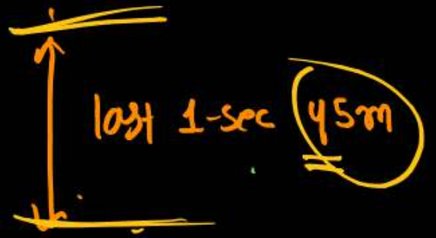
$45 = \frac{10}{2}(2n-1)$

9

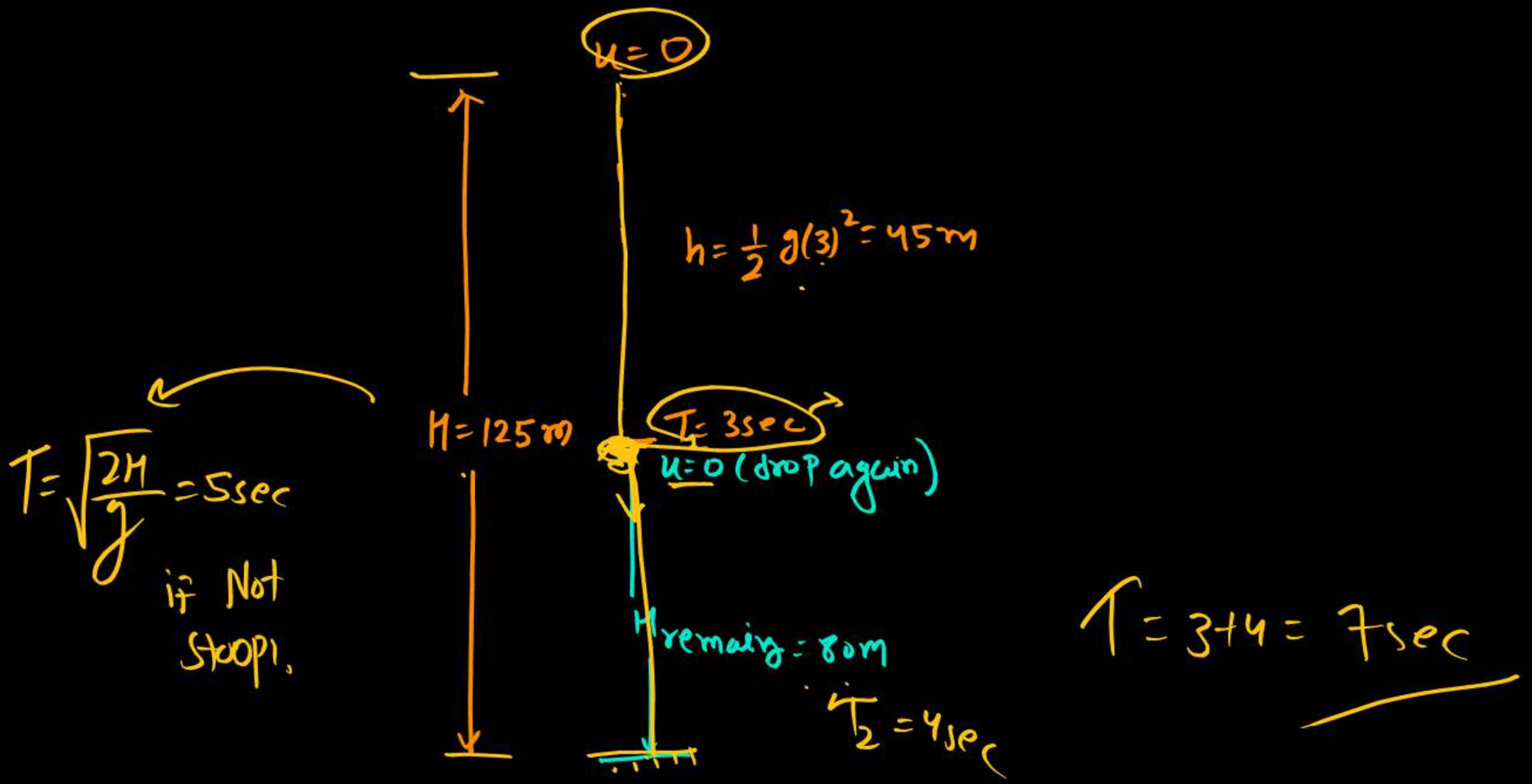
$10 = 2n$

$n = 5$

Total time of flight = 5sec



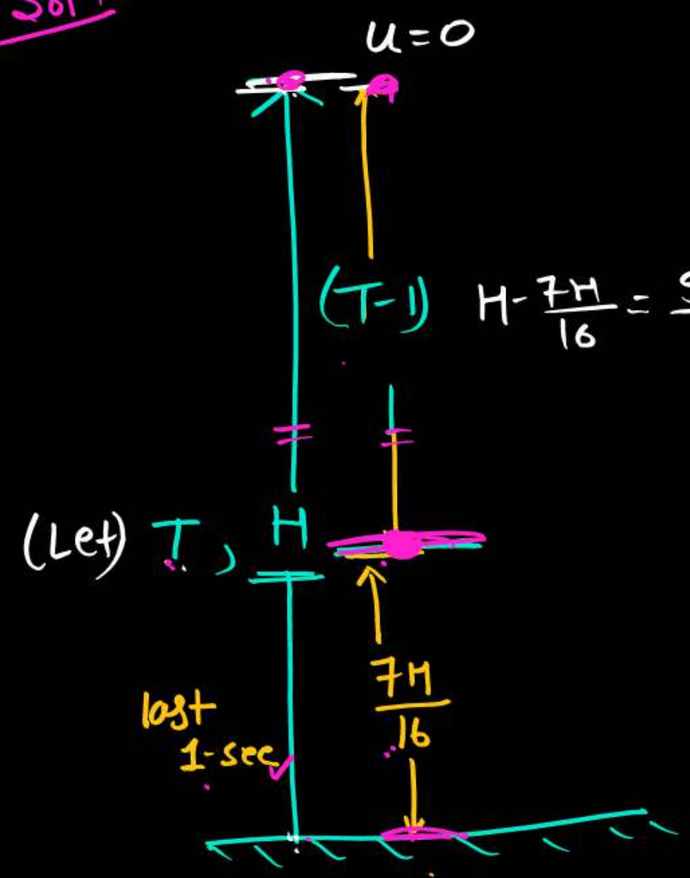
(16) Ball is droped from height 125m after 3sec it stoped and relesed at same instant Find Total time of flight.



$T = 3 + 4 = 7sec$

(17) Ball is droped from Height H and it travell $\frac{7H}{16}$ in last sec of Journey then find 'H'

Soln



Soln

$$H = \frac{1}{2}gT^2$$

$$\frac{9H}{16} = \frac{1}{2}g(T-1)^2$$

$$\frac{16}{9} = \left(\frac{T}{T-1}\right)^2$$

$$\sqrt{\frac{16}{9}} = \frac{4}{3} = \frac{T}{T-1}$$

$4T - 4 = 3T$

$T = 4s$ → $H = \frac{1}{2}g(4)^2 = 80m$

MP*

1-sec Kemtrum.

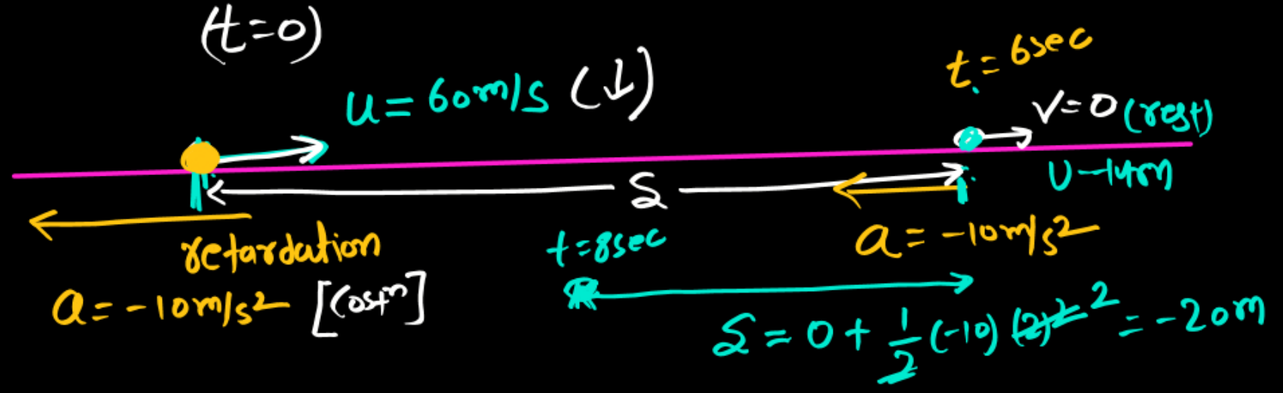
div $\frac{7H}{16} = 5, 15, 25, 35$

$\frac{7H}{16} = 35$ 5

$H = 80m$

Must copy

(18)



(i) find Stopping dist^n $S = \frac{u^2}{2(a)} = \frac{60 \times 60}{2 \times 10} = 180m$

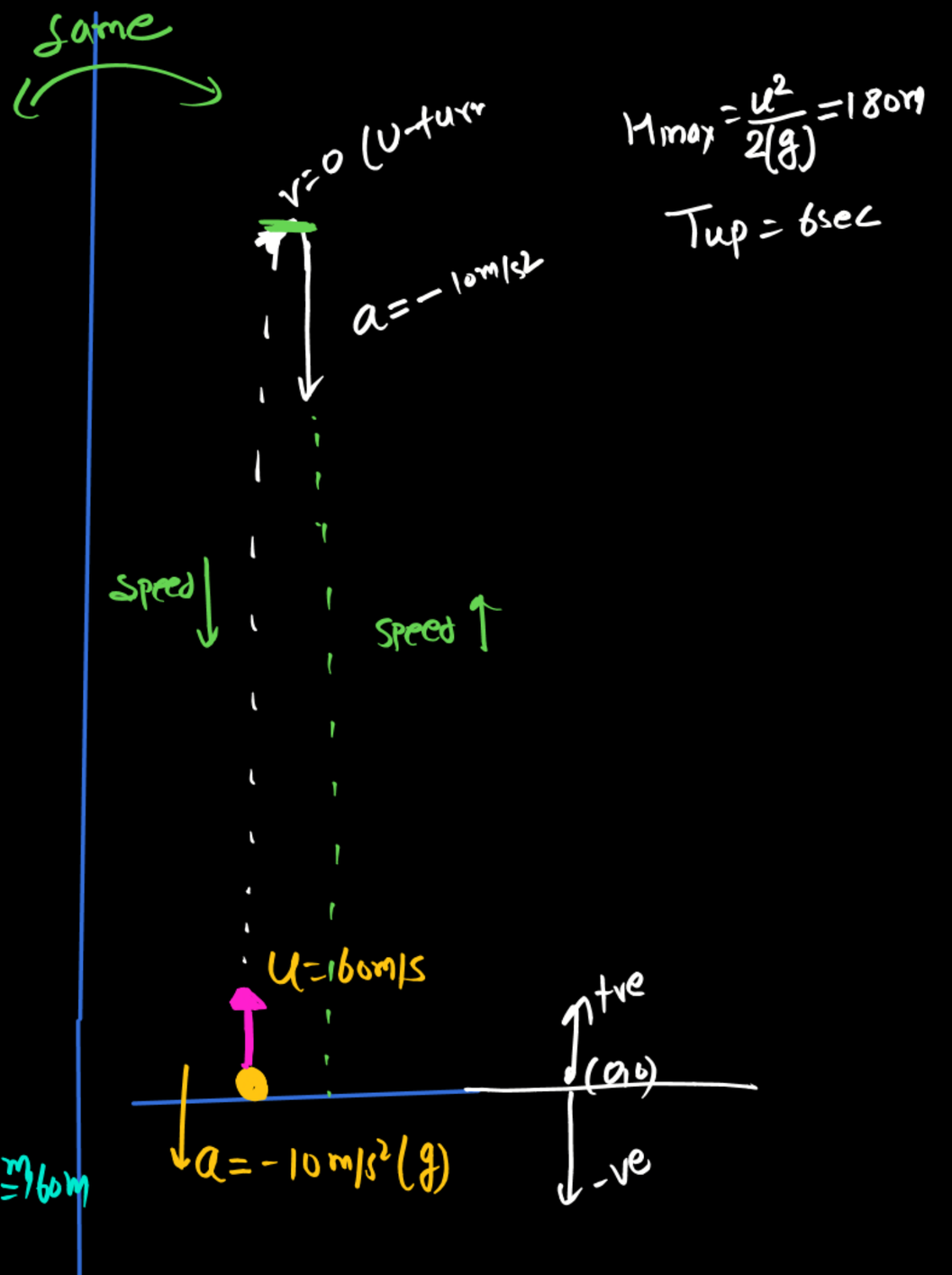
(ii) Stopping time $\Rightarrow v = u + at$

$0 = 60 - 10t$ $\boxed{t = 6sec}$

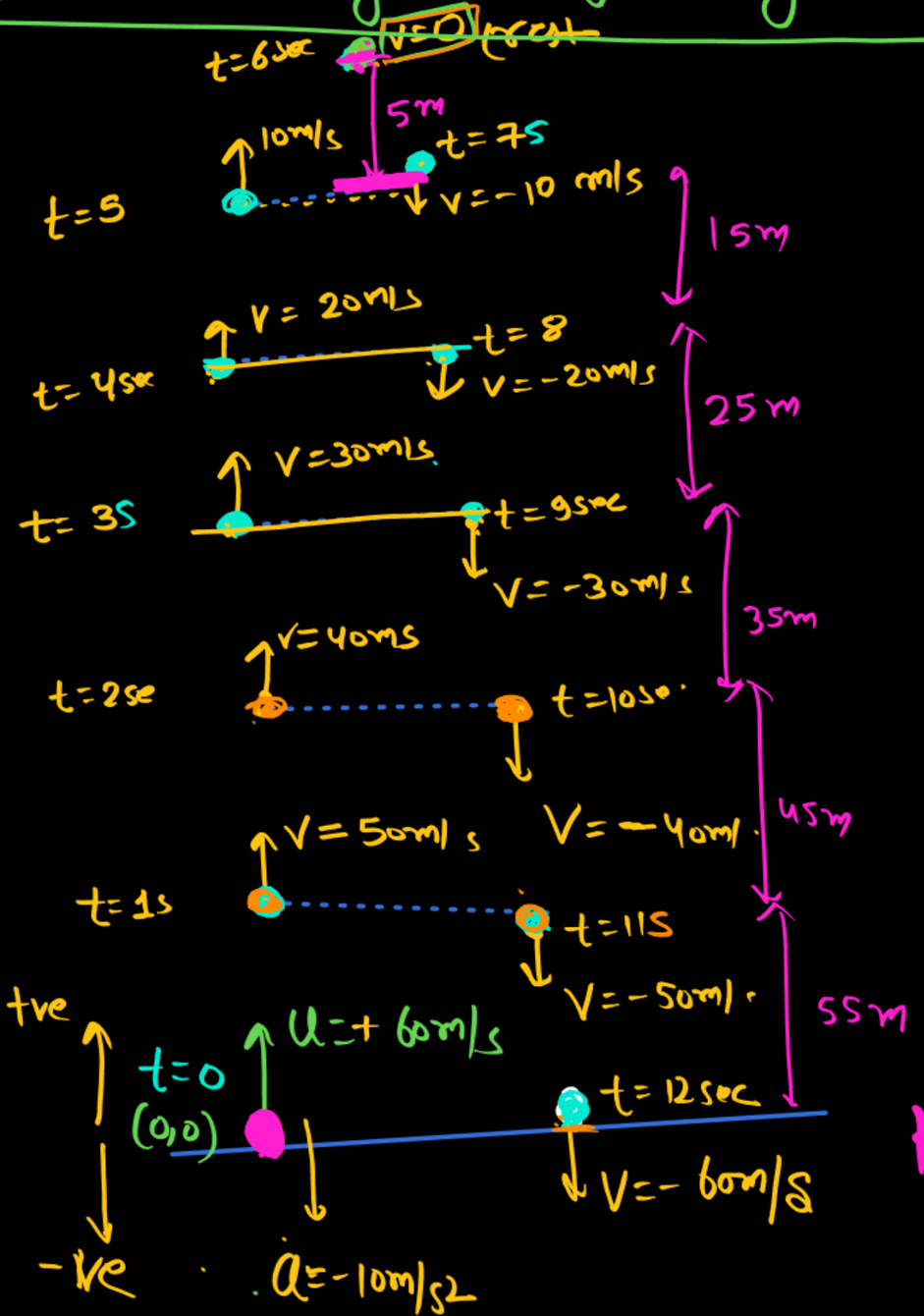
(iii) distance in 8-sec $= 180 + 20 = 200m$

(iv) displacement in 8-sec $\Rightarrow 180 - 20 = 160m$ ✓

$S = ut + \frac{1}{2}at^2 = 60 \times 8 - \frac{1}{2}(10)(8)^2 = 480m - 320m = 160m$



Motion Under gravity from ground to ground



$H = \frac{1}{2} g (6)^2 = 180m$

(i) speed of Projection same as speed of collision

(ii) $\vec{V}_{Projecti} = -\vec{u}_{collision}$

(iii) $T_{up} = \frac{u}{g} = T_{down}$

(iv) $T_{Total} = \frac{2u}{g}$

(v) $H_{max} = \frac{u^2}{2g} = \frac{60 \times 60}{2 \times 10} = 180m$

(vi) Same height ke time ko add kar do total time of flight mil jayga

(v) Avg velocity Point of Projection to maxm height $= \frac{60+0}{2} = 30m/s$

$= \frac{180}{6} = 30m/s$

(vi) $u_{mid\ point\ of\ H_{max}} = \sqrt{\frac{(60)^2+0}{2}} = \frac{60}{\sqrt{2}}$

(viii) Total dist^n = 180+180 m

(vii) disp^m = 0

(19) Ball is projected with speed 40 m/s then find

(i) $H_{max} = 80m$

(ii) $T_f = 8sec$

(iv) $T_{upward} = 4sec$

(v) speed after $t = 5sec, 6sec$ (10 m/s, 20 m/s)

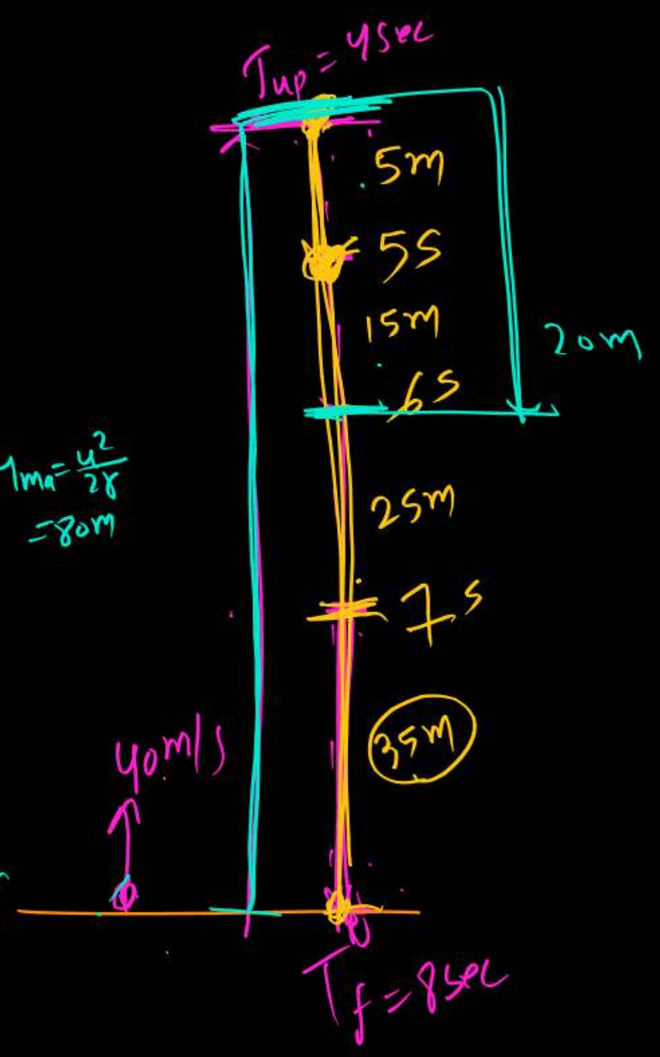
(vi) distance in 6 sec. = 100 m

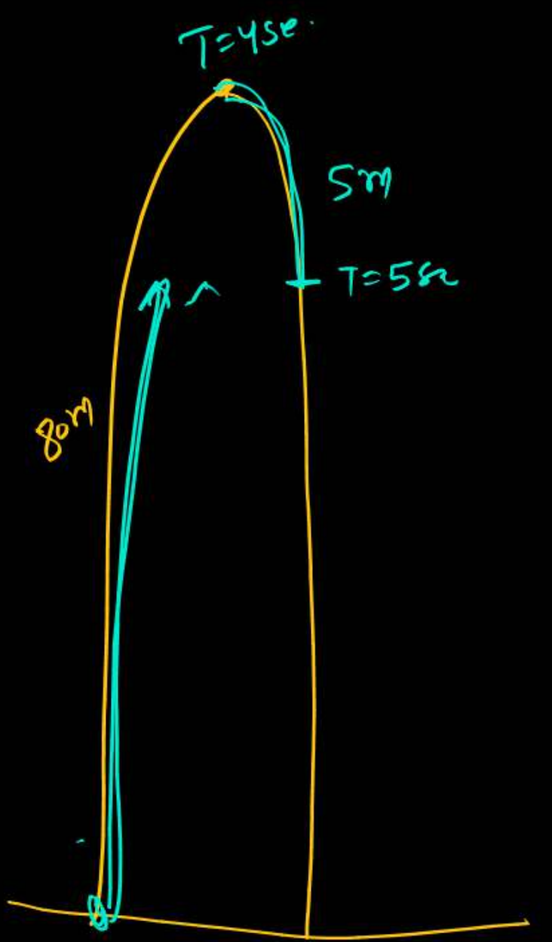
(vii) displacement in 6sec = 60m

(viii) Avg speed in 7sec $= \left(\frac{45+80}{7}\right)$

(ix) Avg velocity in 5sec $= \frac{75}{5}$ ✓

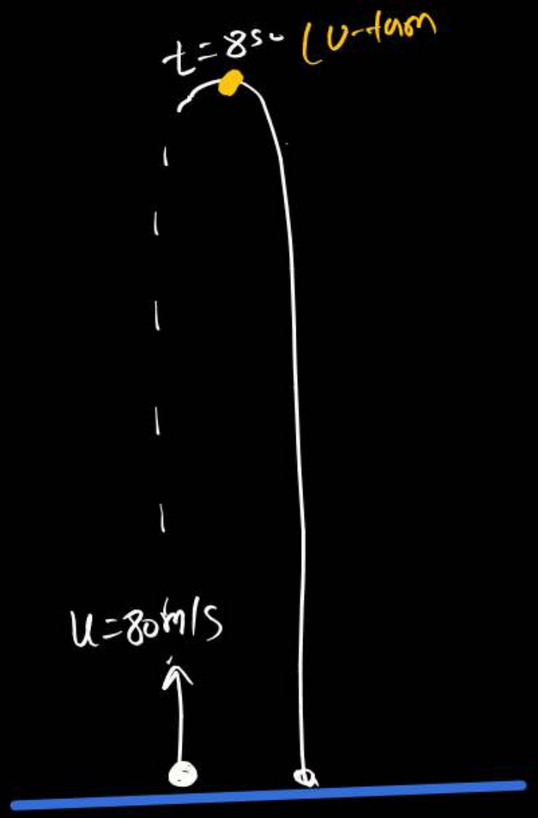
(x) distance moved in 8th sec. = 35m ✓





(20) Object is Bojected with 80m/s then find Avg. speed and Avg velocity in 8-sec -

No Gpr/



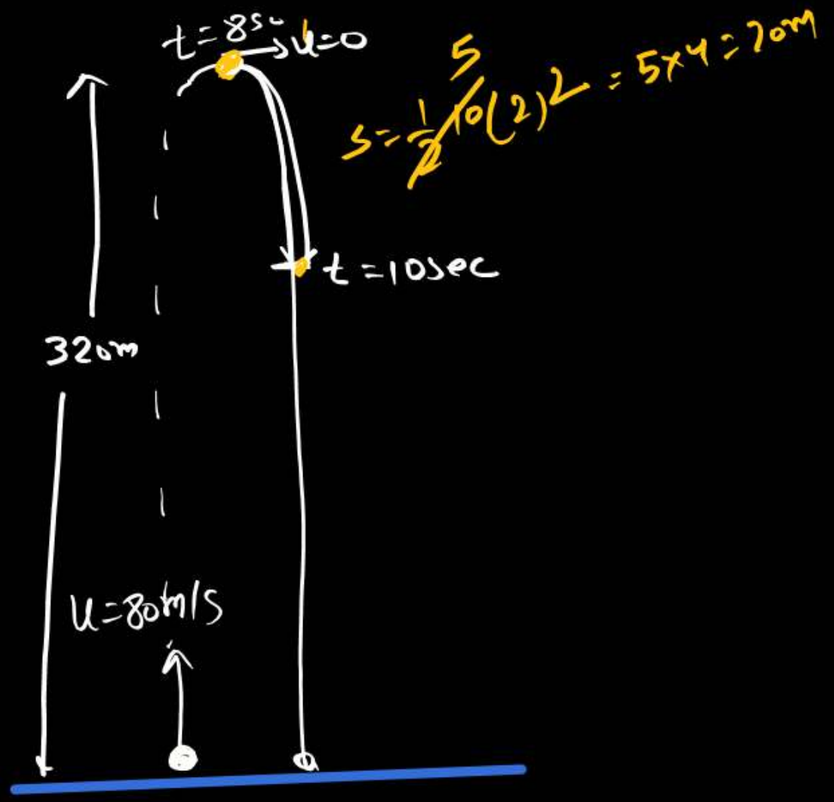
$V_{Avg} = \frac{80+0}{2} = 40 m/s = Ag\ speed$

$H_{max} = \frac{80 \times 80}{2 \times 10} = 320m$ ✓

$V_{Ag} = \frac{320}{8} = 40 m/s$

(20)

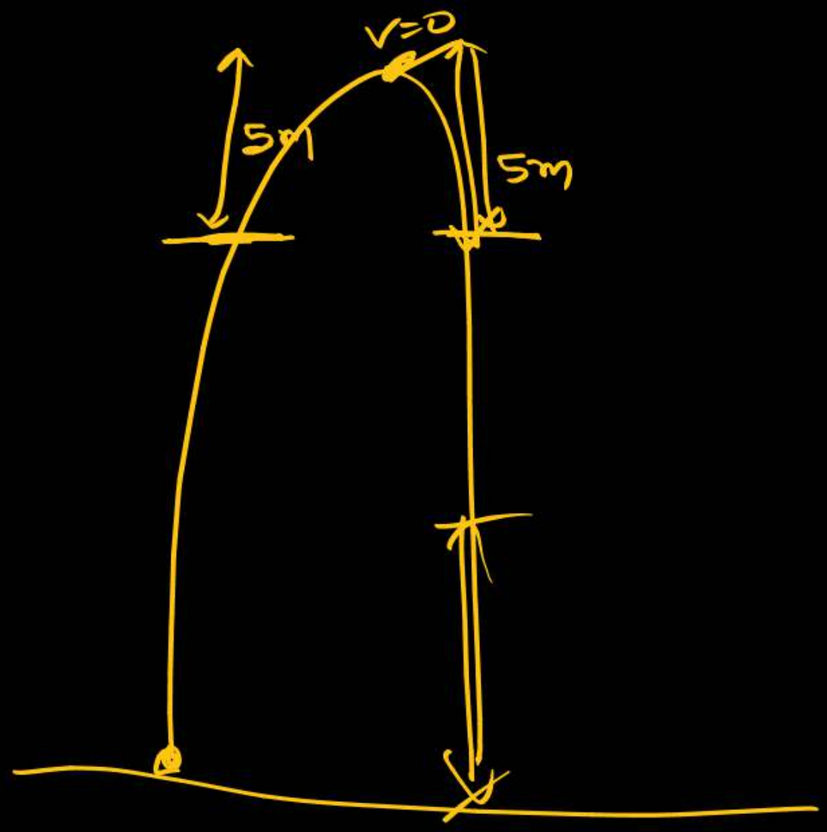
Object is Projected with 80m/s then find Avg. speed and Avg velocity in 10-sec.



$$\text{Avg speed} = \frac{340}{10} = 34\ m/s$$

$$\text{Avg velocity} = \frac{300}{10} = 30\ m/s$$

(21) Two object is projected with 40 m/s and 60 m/s respectively then find ratio of distance in last-sec of upward Journey.



⟶ 1 : 1

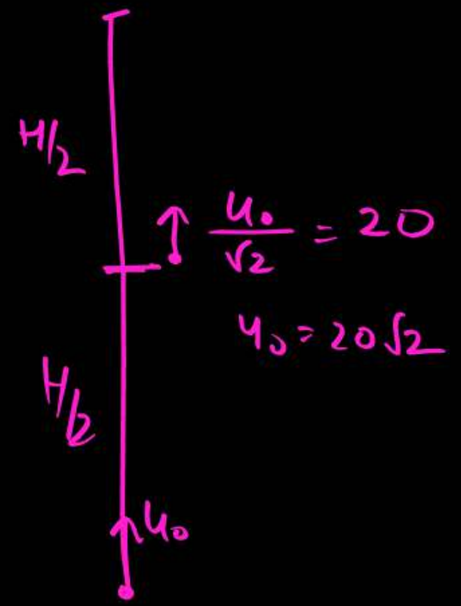
Ans

(22) A Ball is thrown upward with speed 40 m/s then find Avg. velocity of upward Journey. and speed at half of the maximum Height.

done

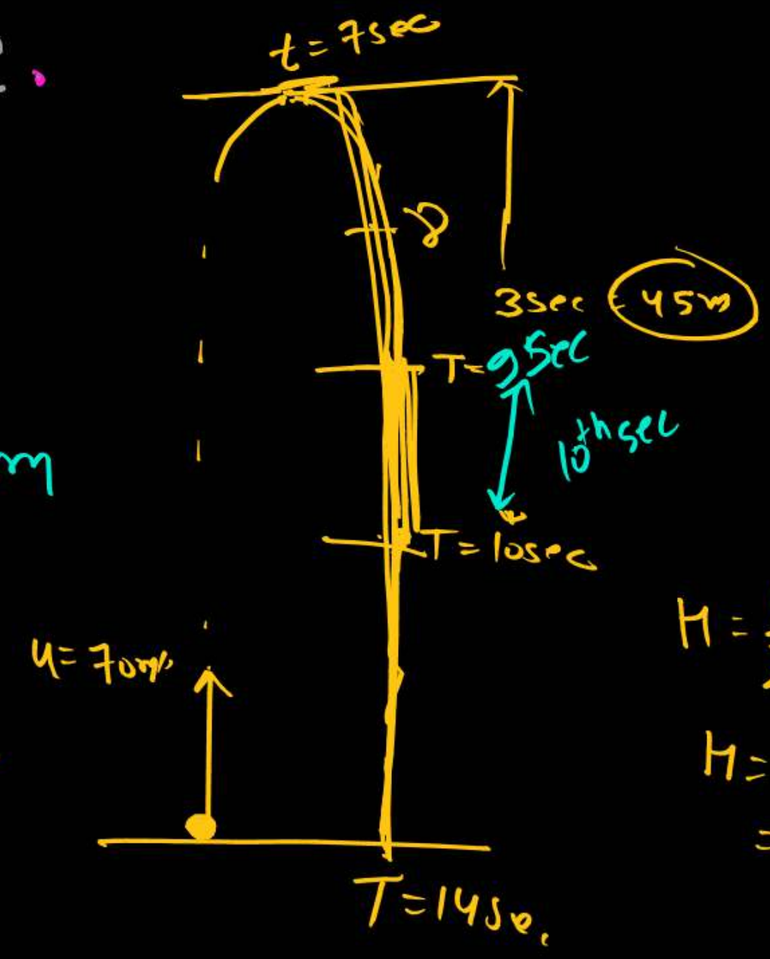
No cpy

23 A Ball is thrown upward with u_0 if its velocity at half of Maximum Height is 20 m/s then find it velocity u_0, max^m Height :-



$$H_m = \frac{u_0^2}{2g} = \frac{20 \times 20 \times 2}{2 \times 10} = 40\,m$$

(24) Ball is projected up with 70m/s then find displacement in 10th sec and 10 sec.



$H = \frac{1}{2}(10)(7)^2$

$H = 5 \times 49$

$= 245m$

disp^m in 10sec $= 245 - 45 = 200m$

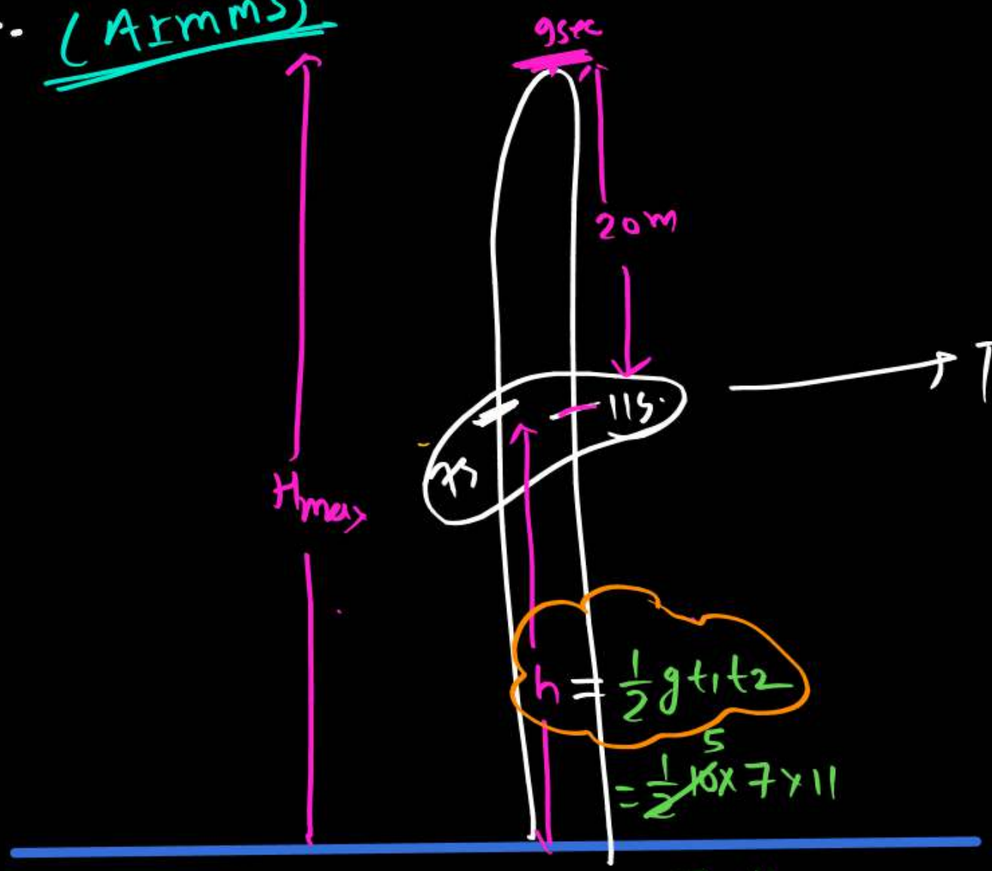
disp^n $= 245 + 45 = 290m$

* disp in 10th sec $= 25m$

distance in 10th sec

(25)

Ball is projected up its position at $t = 7s$ and $t = 11s$ is same then find velocity of projection and max^m Height. (Airmms)



$T_f = 18sec$

$T_{up} = 9sec$

$u = 90m/s$

$h = \frac{1}{2} g t^2 = 405m$

$h = 405 - 20$

$= 385m$

(26)

object is projected up with u its height at 3sec and 13 sec is same find u and that Height.

$T_f = 16s$

$T_{up} = 8s$

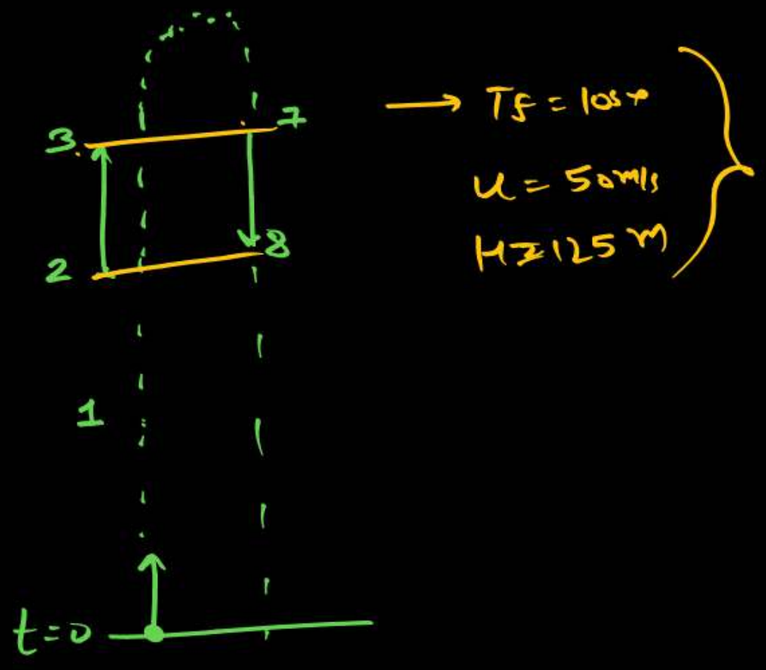
$u = 80 m/s$

$H = 320m$ ✓

(27)

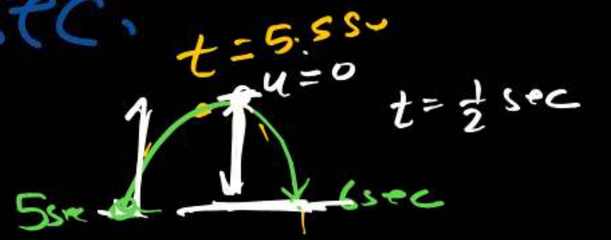
Ball is projected up its distance travelled in 3rd sec and 8th sec is same then find max^m height.

Sol^n



(28)

Ball is projected up with 55 m/s then find distance in 6th sec.

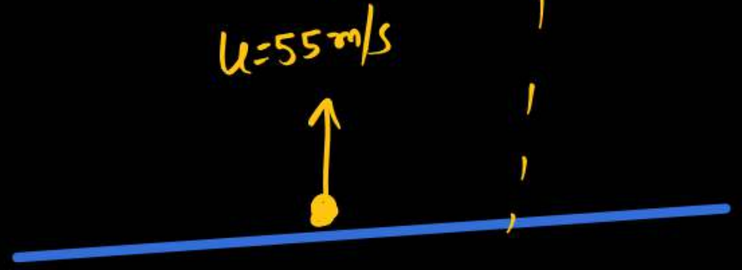


disp^{m} in 6th sec = 0

$h = ut + \frac{1}{2}at^2$

$= 0 + \frac{1}{2} 10 (\frac{1}{2})^2$

$= \frac{10}{8} = 1.25 m$



disp^{m} in 6th sec. 1.25 + 1.25

= 2.5 m

Question

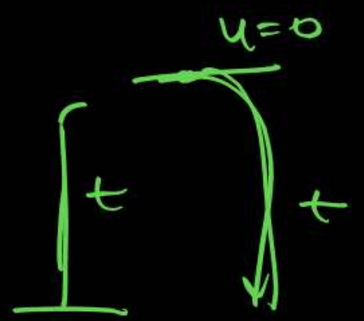
No copy

29

If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is [2003]

upward

1. ut
2. $\frac{1}{2}gt^2$
3. $ut - \frac{1}{2}gt^2$
4. $(u + gt)t$



Question

30 No copy

The ratio of the distance traveled by a freely falling body in the 1st, 2nd, 3rd and 4th second:

[MR* (2022)]

1. 1 : 1 : 1 : 1
2. 1 : 2 : 3 : 4
3. 1 : 4 : 9 : 16
4. 1 : 3 : 9 : 16

(e) 1 : 3 : 5 : 7 : 9 : 11 : 13

Question

(31) no copy

A stone falls freely under gravity. It covers distances h_1, h_2 and h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between h_1, h_2 and h_3 is: **(2013)**

1. $h_1 = h_2 = h_3$
2. $h_1 = 2\,h_2 = 3\,h_3$
3. $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$
4. $h_2 = 3\,h_1$ and $h_3 = 3\,h_2$

Question 32

No copy

A boy standing at the top of a tower of 20 m height drops a stone. Assuming $g = 10\ m/s^2$, the velocity with which it hits the ground is: **(2011 Pre)**

1. 10.0 m/s
2. 20.0 m/s
3. 40.0 m/s
4. 5.0 m/s

$V = \sqrt{2 \times 10 \times 20}$

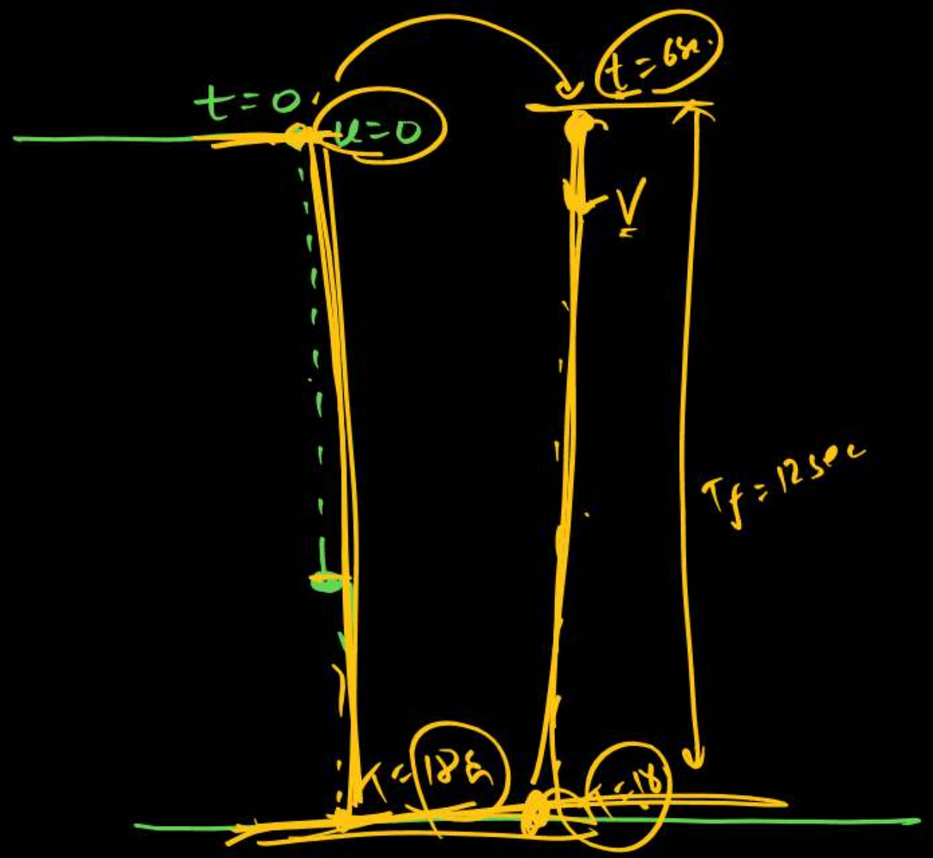
$= \sqrt{400}$

Question 33

copy

A ball is dropped from a high rise platform at $t = 0$ starting from rest. After 6 seconds another ball is thrown downwards from the same platform with a speed v. The two balls meet at $t = 18$ s. What is the value of v? **(2010 Pre)**

1. 75 m/s
2. 55 m/s
3. 40 m/s
4. 60 m/s



$$H = \frac{1}{2}g(18)^2 = v(12) + \frac{1}{2}g(12)^2$$

$$5(18)^2 = 12v + 5(12)^2$$

Unknown

Question 34

No Copy

Two bodies, A (of mass 1 kg) and B (of mass 3 kg) are dropped from heights of 16 m and 25 m, respectively. The ratio of the time taken by them to reach the ground is: (2006)

1. 5/4
2. 8/5
3. 5/8
4. 4/5

$$\frac{t_1}{t_2} = \sqrt{\frac{2h_1}{2h_2}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

Question

The water drop falls at regular intervals from a tap 5 m above the ground. The third drop is leaving the tap at instant the first drop touches the ground. How far above the ground is the second drop at that instant? **(1995)**

1. 3.75 m
2. 4.00 m
3. 1.25 m
4. 2.50 m

done

Question

35 No copy

A particle is thrown vertically upward. Its velocity at half of the height is 10 m/s, then the maximum height attained by it: (g = 10 m/s^2) **(2001)**

1. 8 m
2. 20 m
3. 10 m
4. 16 m

Question 36

A body starts falling from height 'h' and travels distance $h/2$ during last second of motion then time of flight is (in second): **(1999)**

done

1. $\sqrt{2}-1$
2. $2+\sqrt{2}$
3. $\sqrt{2}+\sqrt{2}$
4. $\sqrt{3}+2$

Question 37

No copy

A body dropped from a height h with initial velocity zero, strikes the ground with a velocity 3 m/s. Another body of same mass dropped from the same height h with an initial velocity of 4 m/s. The final velocity of second mass, with which it strikes the ground is:

[MR* (1996)]

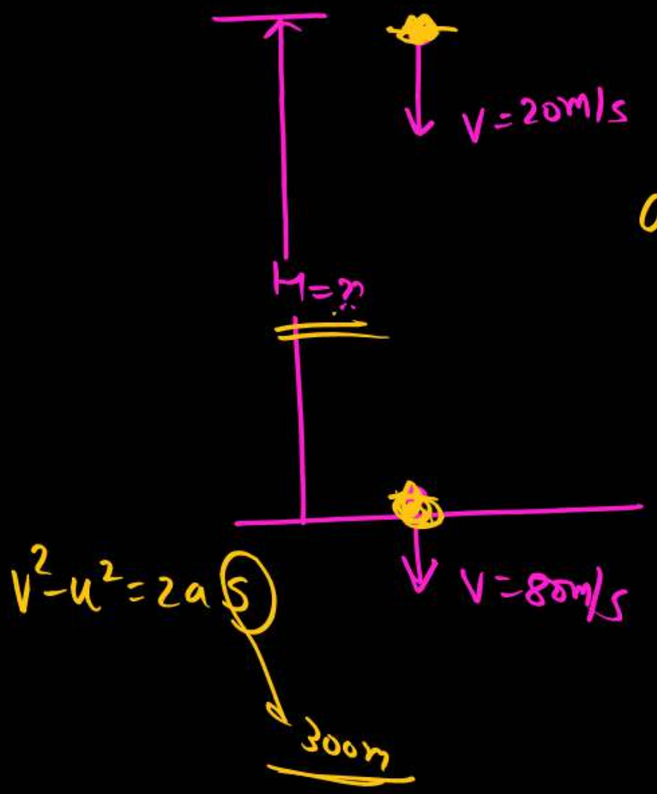
eqn of mol

1. 5 m/s
2. 12 m/s
3. 3 m/s
4. 4 m/s

Question 38

A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is: (g = 10 m/s^2) [2020]

1. 340 m
2. 320 m
3. 300 m
4. 360 m



$v - u = at$

$80 - 20 = 10t$

$60 = 10t$

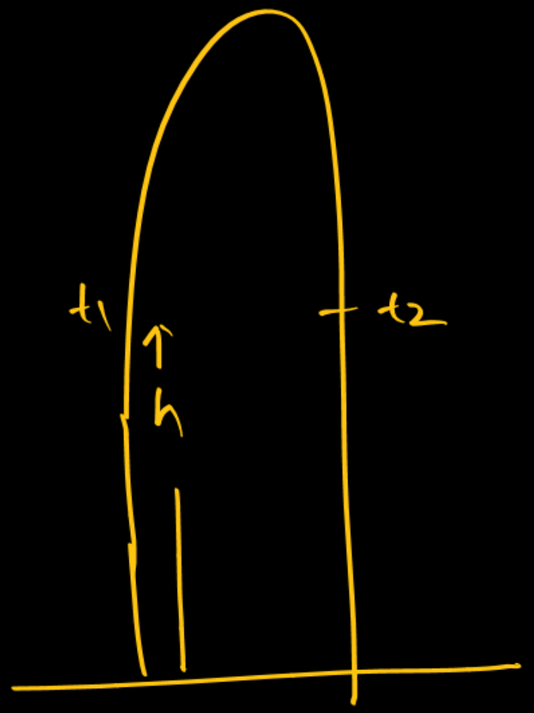
$t = 6s$

$H = \left(\frac{u+v}{2}\right)t = \left(\frac{20+80}{2}\right)6$

$= 50 \times 6$

$= 300m$

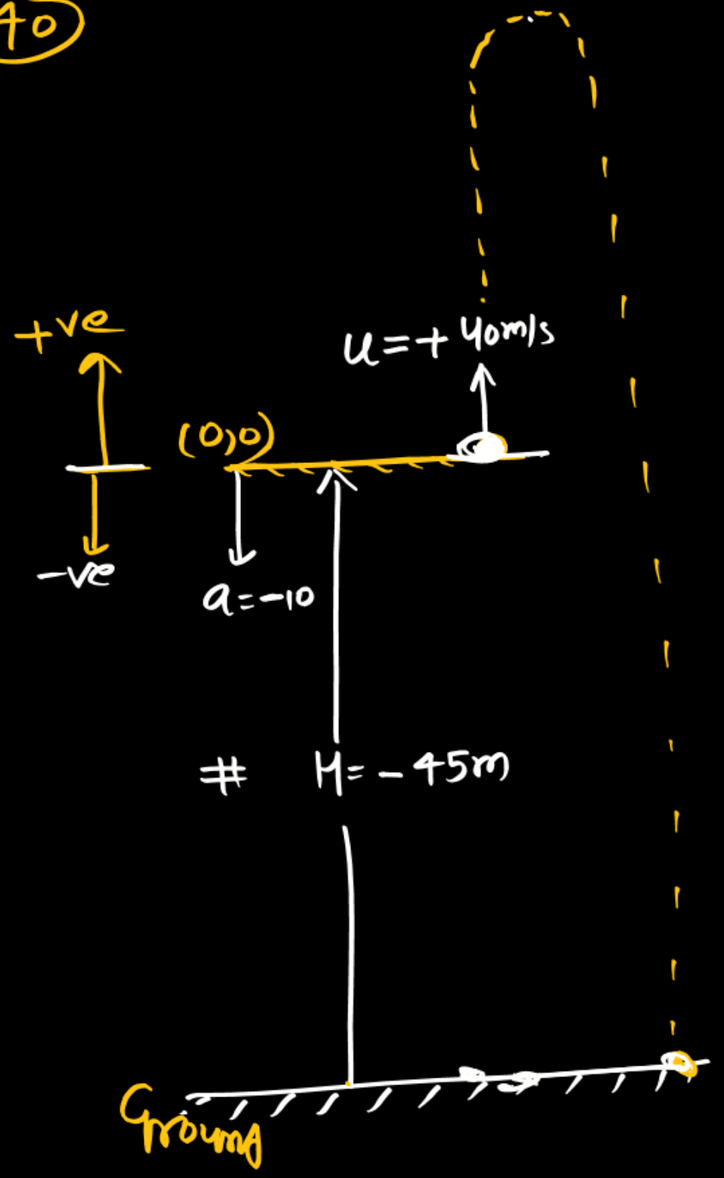
(39) Object is Project ⇒ cross same height at time t_1 & t_2 then find that height.



$$h = \frac{1}{2} g t_1 t_2 \checkmark$$

Find Time of Flight:

(40)



$$s = ut + \frac{1}{2}at^2$$

$$-45 = 40t + \frac{1}{2}(-10)t^2$$

$$5t^2 - 40t - 45 = 0$$

$$t^2 - 8t - 9 = 0$$

$$t^2 - 9t + t - 9 = 0$$

$$t(t-9) + 1(t-9) = 0$$

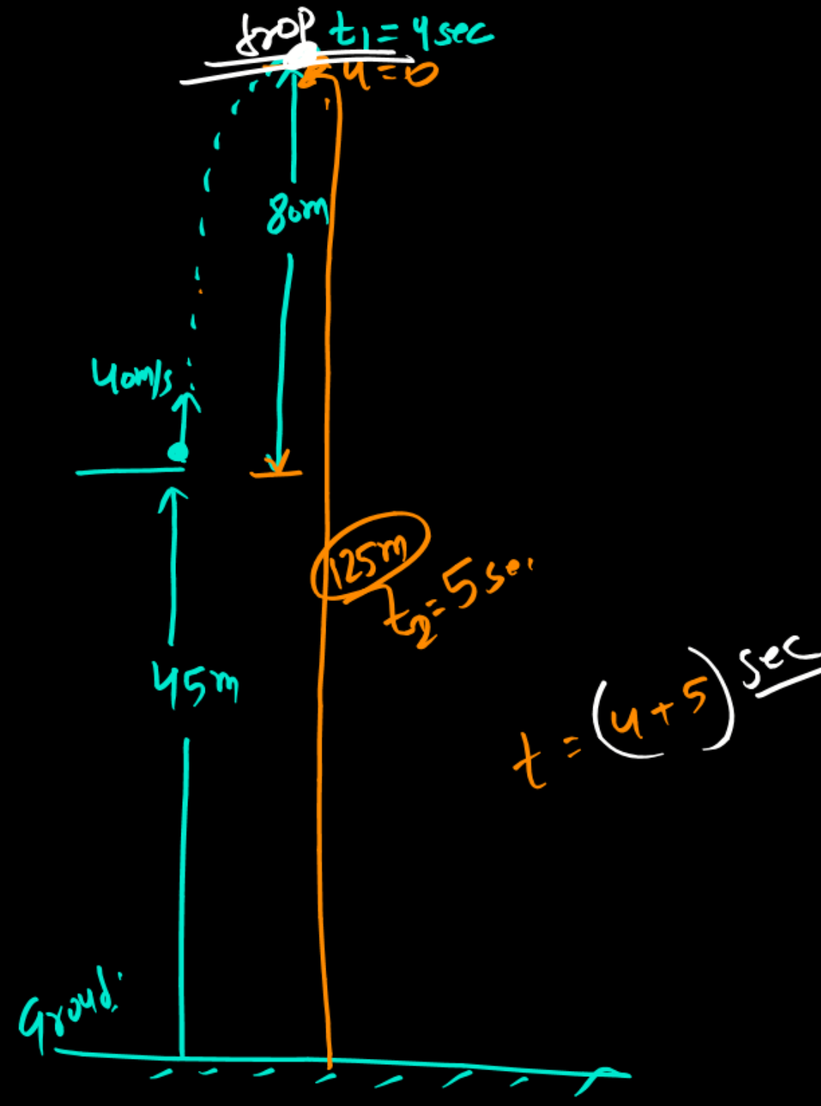
$$(t-9)(t+1) = 0$$

$t_1 = 9$ ✓

$t_2 = -1$ sec ✗

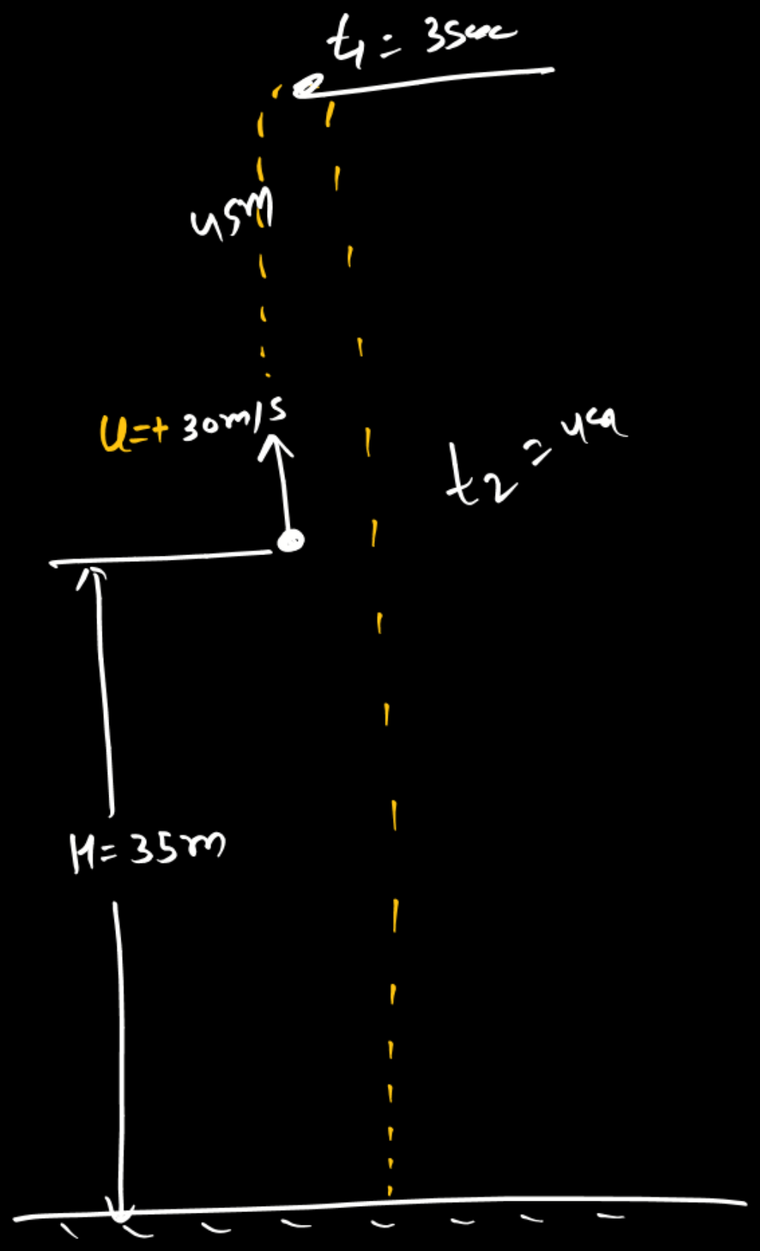
Method-1

MR*



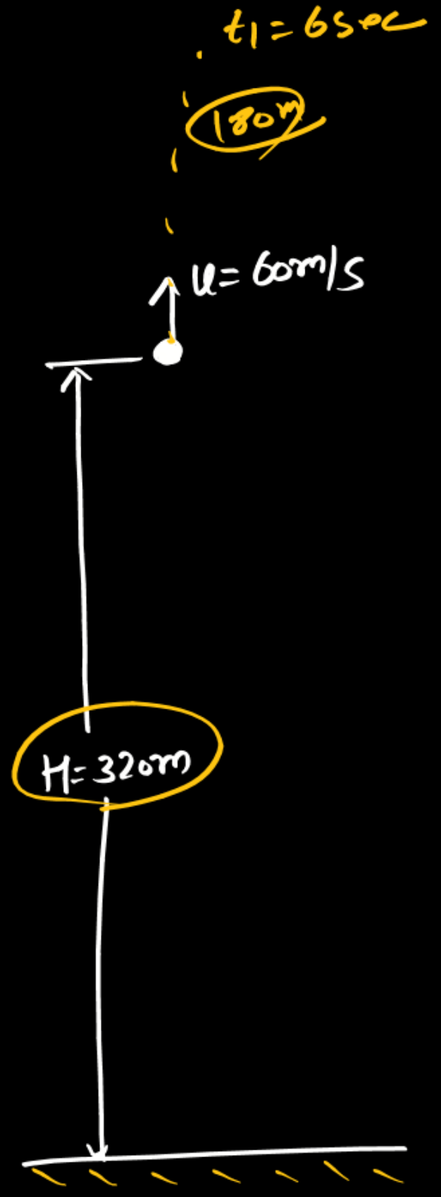
$t = (4+5)$ sec

(41)



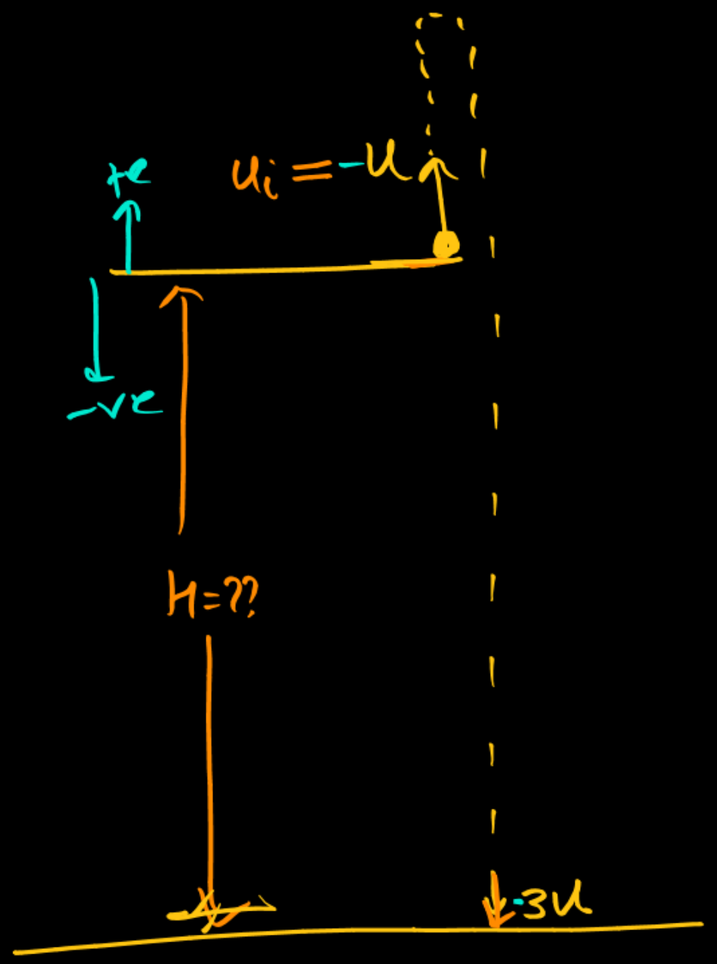
$T = 7\text{sec}$ ✓

(42)



$T_f = 16$ sec

(43) find Time of flight & maximum height of Tower



$$-3u = u - gt$$

$$gt = 4u$$

$$t = \frac{4u}{g}$$

$V = u + at$

$3u = u - gt$

$\frac{2u}{g} = t$

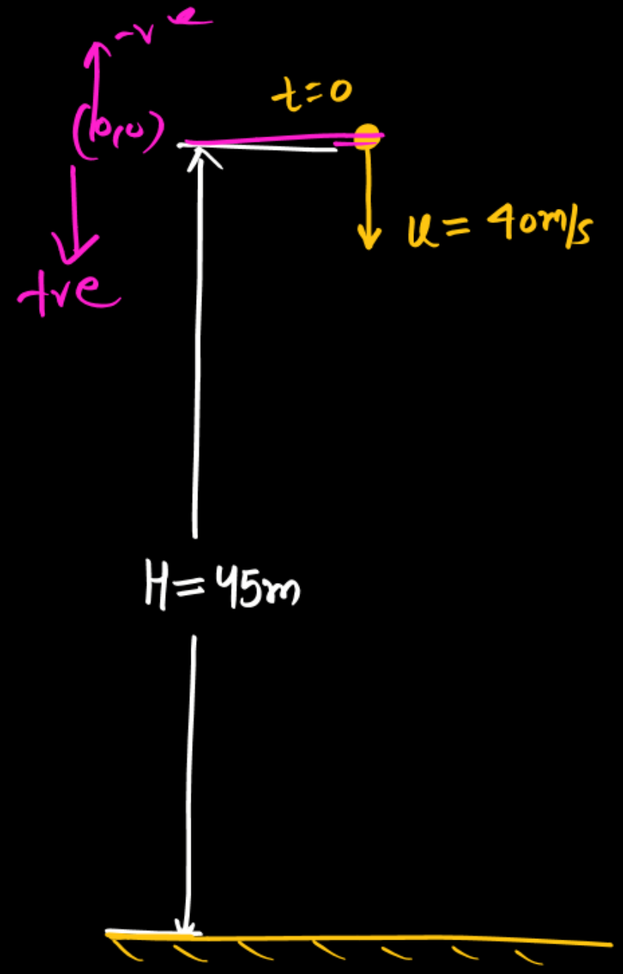
+

3rd eqⁿ

$$(-3u)^2 - (u)^2 = 2(-g)H$$

$$H = \frac{4u^2}{g}$$

44



$s = +45m$ ✓

$u = +40m/s$ ✓

$a = +10m/s^2$ ✓

eqn of motn.

$$s = ut + \frac{1}{2}at^2$$

$$45 = 40t + \frac{1}{2}(10)t^2$$

$$5t^2 + 40t - 45 = 0$$

$$t^2 + 8t - 9 = 0$$

$$t^2 + 9t - t - 9 = 0$$

$t_1 = -5$ ✗

$t_2 = +1s$ ✓

∴

drop $t=0$

80m

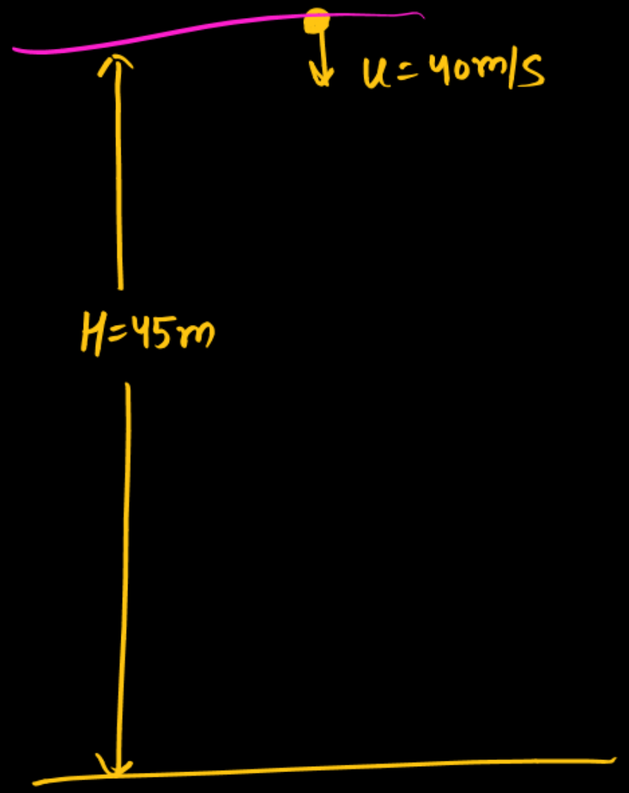
$V = 40m/s$ $(T = 4sec)$

45m

$T_{(next)} = 1sec$

$H = 125m$

$T = 5s$


u = 40m/s
H = 45m

(Q5) Find time of fly



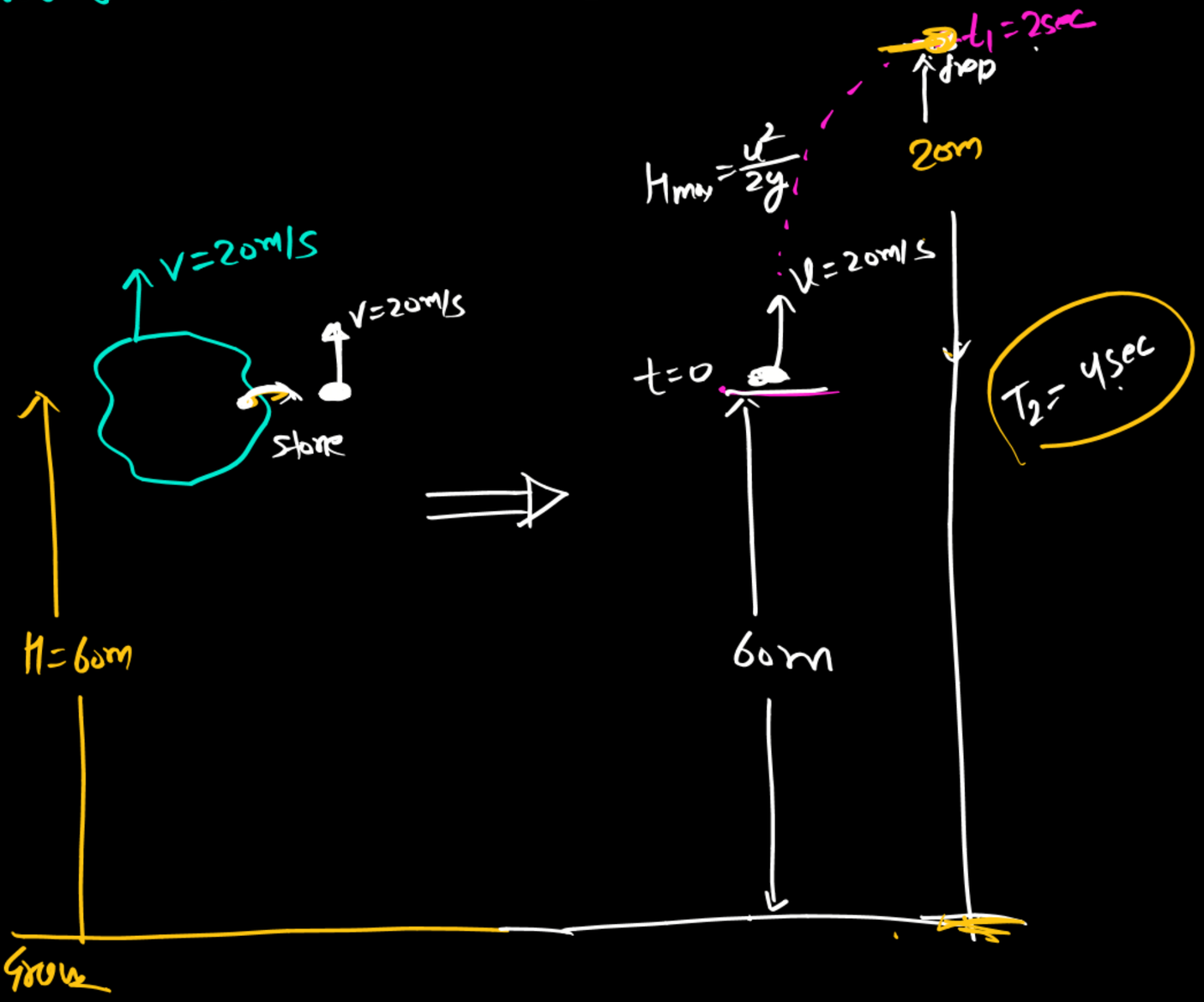
$t_1 = 2\,sec$

$S = ut + \frac{1}{2}at^2$

MR^X Box →

Jab bhi object ko moving frame se release (drop) karte hai, object frame ka velocity share kar leta hai [due to inertia] but acceleration nahi leta.

(47) Ballon is moving up with constant velocity 20m/s at height 60m stone is released from Ballon, find Time of flight & max height attain by stone.

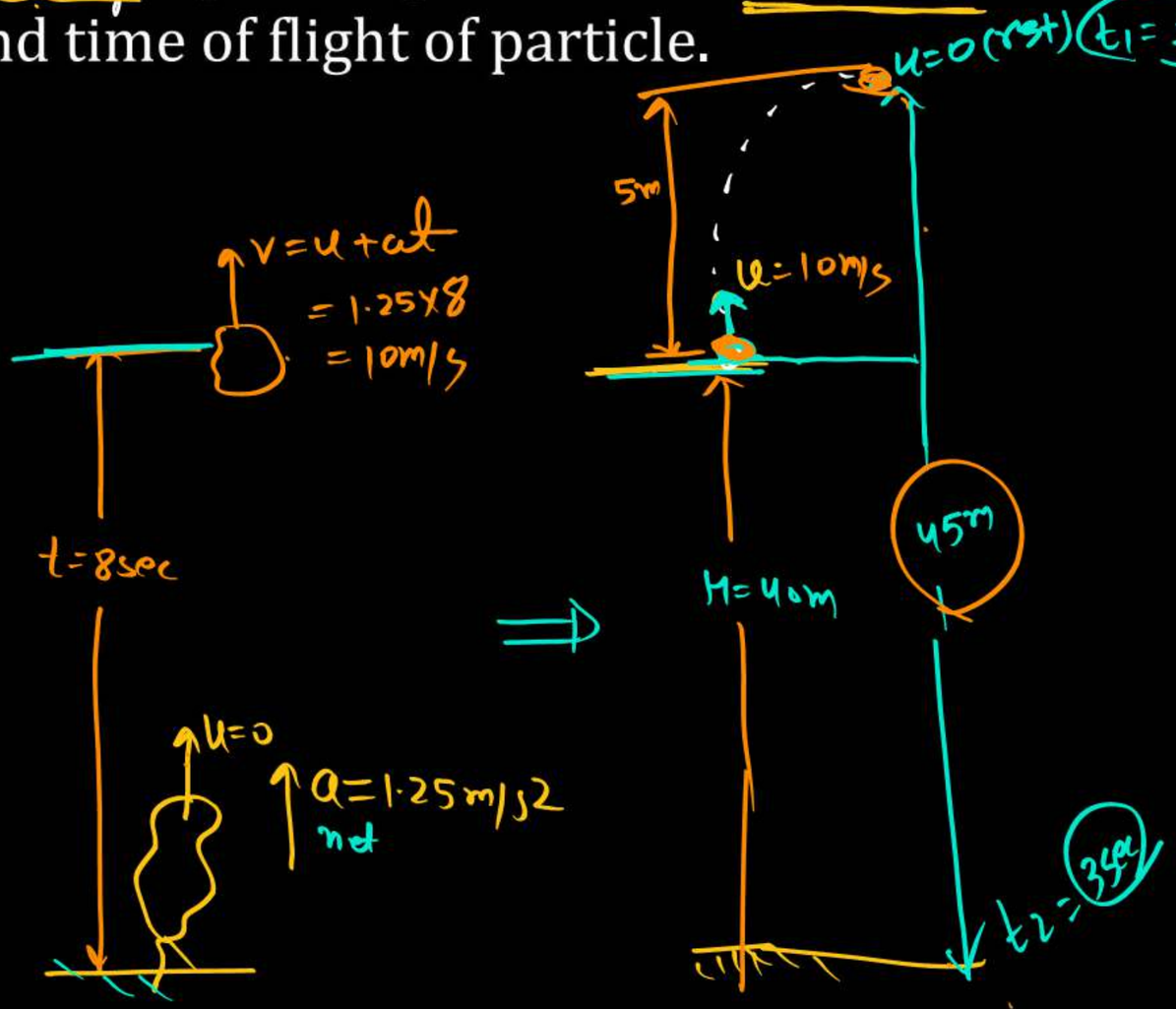


$T = 6\,sec$ Ans

$H_{max} = 80m$ ✓
from g

Question

A balloon starts raising up from ground with 1.25 m/s² after 8 sec a small particle is dropped then find time of flight of particle.



$h = ut + \frac{1}{2}at^2$

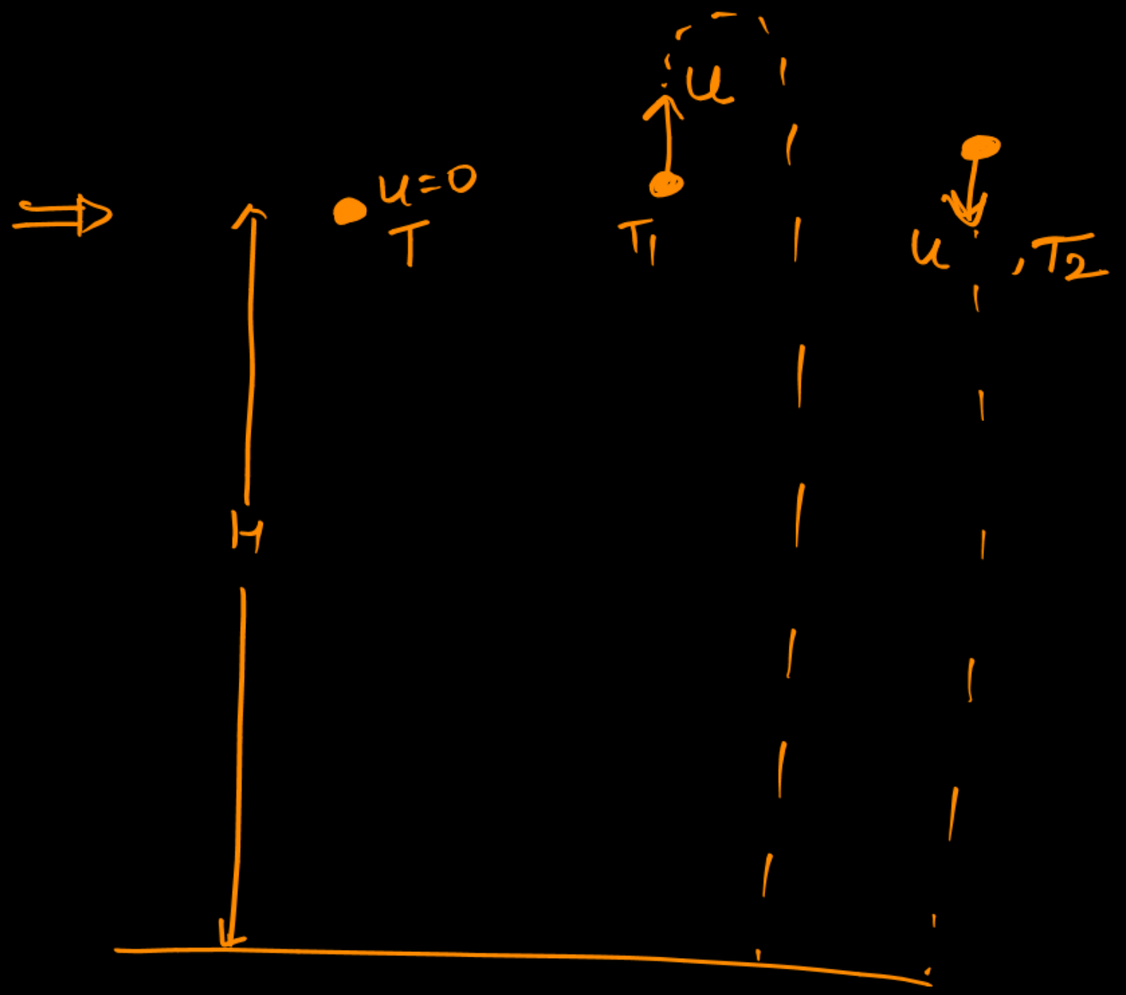
$= 0 + \frac{1}{2}\frac{5}{4}(8)^2$

$= \frac{5 \times 8 \times 8}{8}$

$= 40m$

$T_f = 3 + 1 = 4sec$

Jugglar wants to keep n - Ball in air, if he throw each Ball with speed u, then find time intaval b/w each Ball.



$$T = \sqrt{T_1 T_2}$$

Try to prove it //

MR Sagar

(Position) $\vec{r} = t\hat{i} + 4t\hat{j}$

(a) 1-D (12.1) (+1)

(b) 2-D (60%)

(c) Can't say (20%)

Jugglar wants to keep 5 - Ball in air, with time intraval 2-sec then find intial speed of Protection ??

Question

Rocket starts his motion in upward direction with acceleration 10 m/s^2 upward. After 5 sec engine off then find maximum height from ground.

Question

A man throws ball with the same speed vertically upwards one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time? (Given $g = 9.8\ m/s^2$) **[MR* (2003)]**

1. More than 19.6 m/s
2. At least 9.8 m/s
3. Any speed less than 19.6 m/s
4. Only with speed 19.6 m/s

Question

A person sitting in the ground floor of a building notices through the window, of height 1.5 m, a ball dropped from the roof of the building crosses the window in 0.1 s. What is the velocity of the ball when it is at the topmost point of the window? (g = 10 m/s^2)

(2020-Covid)

1. 14.5 m/s
2. 4.5 m/s
3. 20 m/s
4. 15.5 m/s

Thank

You